



Computer Science Year 2

Algorithms & Data

Estimation, Regression, Classification

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🔥 Last time ...

- Least squares (LS) estimation
 - Minimizes sum of squares between measurements and a model
 - Generally applicable estimator as no assumption is made about the data
 - Best for linear models
- Method of Moments (MoM)
 - Based on equating sample and population moments
 - Simplest estimation approach, intuitive, works well in straightforward cases
 - Not always leading to good results, especially in small sample sizes



Objectives

- Bayesian Estimation
 - Motivation
 - The Bayesian paradigm
 - The MMSE estimator
 - The MAE estimator
 - The MAP estimator
 - Examples



🔥 Classical vs Bayesian estimation

- Classical methods
 - The assumptions leading to asymptotic results may not apply sometimes;
 - Asymptotic approximations are not always reliable, even for medium sample sizes. For small sample sizes, estimators like the MLE (asymptotically justified) can even lead to absurd results;
 - Frequentist estimators work well *on average*, but not necessarily for the data at hand;
 - They are not able to account for any kind of extra-information that may be available;
 - Classical approach to estimation assumes that the parameter to be determined is a deterministic but unknown constant.

🔥 Classical vs Bayesian estimation

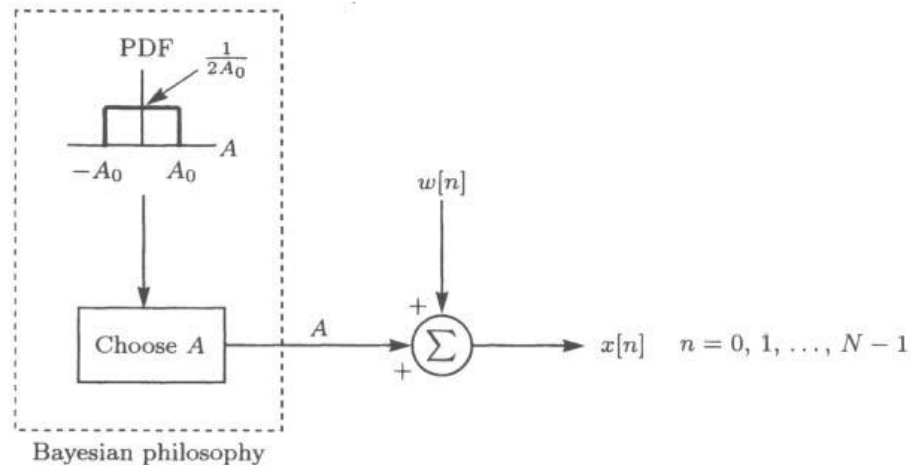
- Bayesian methods
 - In Bayesian approach the unknown parameter is assumed to be a random variable;
 - They enable prior information about the parameters to be incorporated in the estimation procedure;
 - They do not need to be justified by any asymptotic approximation;
 - Bayesian techniques are based on modelling the uncertainty with respect to the parameter θ through a probability distribution.

🔥 The Bayesian MSE

- Remember the DC level in WGN example:

$$x[n] = A + w[n], \text{ where } n = 0, 1, \dots, N-1 \text{ and } w[n] \sim N(0, \sigma^2)$$

- The MVUE of A was found to be the sample mean, assuming $-\infty < A < \infty$ (deterministic unknown) ...
- However, by assigning a particular PDF to the *random variable* (!) A :



- We can attempt to find an estimator of A that would minimize the MSE:

$$B_{MSE}(\hat{A}) = E \left[(A - \hat{A})^2 \right]$$

🔥 The Bayesian MSE

- Classical MSE:

$$mse(\hat{A}) = \int (\hat{A} - A)^2 p(x; A) dx$$

- Bayesian MSE:

$$Bmse(\hat{A}) = \iint (A - \hat{A})^2 p(x, A) dx dA$$

- Whereas the classical MSE depends on A (and hence estimators that attempt to minimize it will usually depend on A), the Bayesian MSE does not! That's because the parameter dependence is integrated away!

🔥 Elements of Bayesian analysis

- **Bayes rule:**
$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

where: $p(\theta|x)$ – posterior $p(x|\theta)$ – likelihood $p(\theta)$ – prior
 $p(x)$ – evidence

- **Definition: Bayesian statistical model**

A statistical model composed of a data generation model, $p(x|\theta)$, and a prior distribution on the parameters, $p(\theta)$.

- Joint distribution:
$$p(x, \theta) = p(x|\theta)p(\theta)$$
- Marginal distributions:
$$p(x) = \int p(x|\theta) p(\theta) d\theta$$
$$p(\theta) = \int p(x|\theta) p(\theta) dx$$

🔥 Example: DC level in WGN (continued)

$$Bmse(\hat{A}) = \iint (A - \hat{A})^2 p(x, A) dx dA$$

$$p(x, A) = p(A|x)p(x)$$

$$Bmse(\hat{A}) = \int \left[\int (A - \hat{A})^2 p(A|x) dA \right] p(x) dx$$

- The Bayesian MSE will be minimized if the integral in brackets can be minimized for each x .
- Taking the derivative:
$$\begin{aligned} \frac{\partial}{\partial \hat{A}} \int (A - \hat{A})^2 p(A|x) dA &= \int \frac{\partial}{\partial \hat{A}} (A - \hat{A})^2 p(A|x) dA \\ &= \int -2(A - \hat{A}) p(A|x) dA \\ &= -2 \int A p(A|x) dA + 2\hat{A} \int p(A|x) dA \end{aligned}$$

🔥 Example: DC level in WGN (continued)

- Setting to zero

$$-2 \int A p(A|x) dA + 2\hat{A} \int p(A|x) dA = 0$$

- And since the conditional PDF must integrate to 1

$$\hat{A} = \int A p(A|x) dA$$

- Finally

$$\hat{A} = E(A|x)$$

🔥 Bayesian estimators

- In general, a Bayesian estimator minimizes the conditional risk, which is the loss (cost function) averaged over the conditional (posterior) distribution of θ , given the observation (measurement) x :

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int C[\theta, \hat{\theta}(x)] p(\theta|x) d\theta$$

- Definition: *The Bayes risk R is the average cost $E[C(\varepsilon)]$ and measures the performance of a given estimator.*

$$R = E[C(\varepsilon)]$$

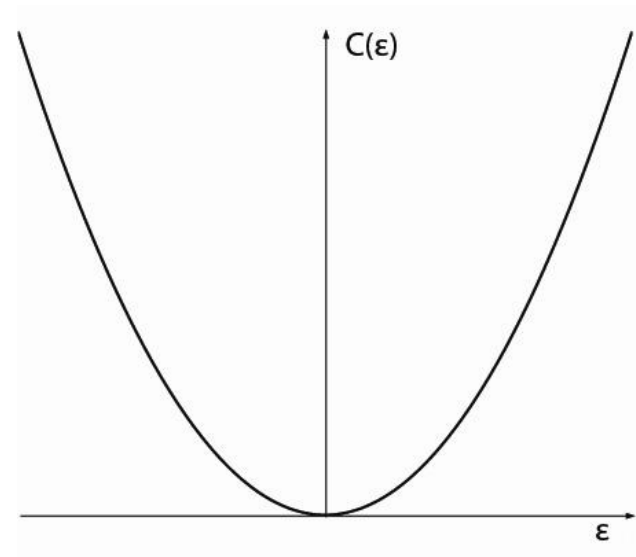
🔥 The Minimum Mean Square Error (MMSE) estimator

- Quadratic error cost function

$$C[\theta, \hat{\theta}(x)] = C(\varepsilon) = \varepsilon^2$$

- The corresponding optimal estimator is the *mean of the posterior PDF*

$$\hat{\theta} = \int \theta p(\theta|x) d\theta = E(\theta|x)$$



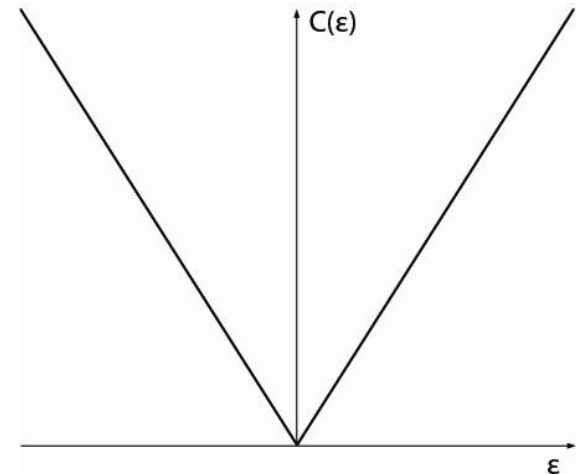
🔥 The Minimum Absolute Error (MAE) estimator

- Absolute error cost function

$$C[\theta, \hat{\theta}(x)] = C(\varepsilon) = |\varepsilon|$$

- General Bayesian estimator

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int C[\theta, \hat{\theta}(x)] p(\theta|x) d\theta$$



- Using the two equations above, the MAE is obtained as

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int |\theta - \hat{\theta}| p(\theta|x) d\theta$$

🔥 The MAE estimator

- The integral can be split into

$$g(\hat{\theta}) = \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) p(\theta|x) d\theta + \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) p(\theta|x) d\theta$$

- In order to differentiate one can use Leibnitz's rule yielding

$$\frac{dg(\hat{\theta})}{d\hat{\theta}} = \int_{-\infty}^{\hat{\theta}} p(\theta|x) d\theta - \int_{\hat{\theta}}^{\infty} p(\theta|x) d\theta$$

- And setting to 0 :-

$$\int_{-\infty}^{\hat{\theta}} p(\theta|x) d\theta = \int_{\hat{\theta}}^{\infty} p(\theta|x) d\theta$$

that is by definition the *median of the posterior PDF*.

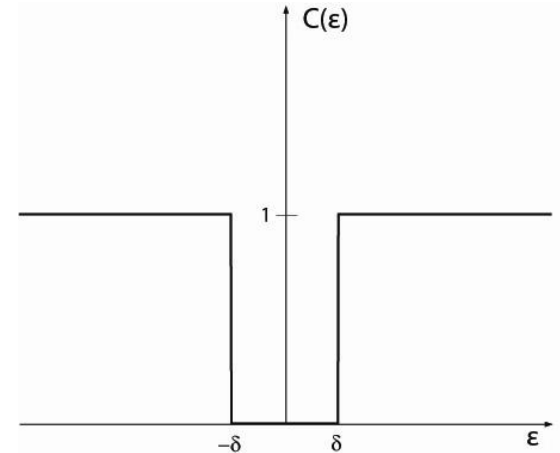
🔥 The Maximum a Posteriori (MAP) Estimator

- Hit-or-miss cost function

$$C(\varepsilon) = \begin{cases} 0, & |\theta - \hat{\theta}| < \delta \\ 1, & \text{otherwise} \end{cases}$$

- General Bayesian estimator

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int C[\theta, \hat{\theta}(x)] p(\theta|x) d\theta$$



- Using the two equations above, the MAP is obtained as

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \int_{|\theta - \hat{\theta}| \geq \delta} p(\theta|x) d\theta$$

🔥 The MAP Estimator

- Or

$$\hat{\theta}(x) = \operatorname{argmin}_{\theta} \left[1 - \int_{|\theta - \hat{\theta}| < \delta} p(\theta|x) d\theta \right]$$

- In order to minimize the expected cost, when $\delta \rightarrow 0$ one should select (the MAP equation)

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} p(\theta|x)$$

that is, the mode of the posterior pdf.

- Using Bayes theorem together with the last equation, we can also write the MAP equation as (more useful in practice)

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} p(x|\theta)p(\theta)$$

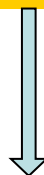
🔥 Example

- Assume that $p(x|\theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right)$

and that the prior pdf is $p(\theta) = \frac{\gamma}{\pi(\theta^2 + \gamma^2)}$

- The MAP estimator can be found as follows:-

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} [\ln p(x|\theta) + \ln p(\theta)]$$



$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} \left[-\frac{(x-\theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right]$$

✿ Example (continued)

$$\hat{\theta}(x) = \operatorname{argmax}_{\theta} \left[-\frac{(x - \theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right]$$

- Differentiating with respect to θ

$$\frac{d}{d\theta} \left[-\frac{(x - \theta)^2}{2\sigma^2} + \ln \frac{\gamma}{\pi(\theta^2 + \gamma^2)} \right] = \frac{x - \theta}{\sigma^2} - \frac{2\theta}{\theta^2 + \gamma^2}$$

- Setting equal to 0 yields

$$\frac{x - \theta}{\sigma^2} = \frac{2\theta}{\theta^2 + \gamma^2}$$

- Finally, rearranging

$$\theta^3 - x\theta^2 + (\gamma^2 + \sigma^2)\theta - \gamma^2x = 0$$



Example: MAP "Processor" I/O Curves

